

MovieLens Data Analysis: Feature Engineering & Exploration Report

By

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HNG Stage 1 Task

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Executive Summary

This report presents an analysis of the Movielens dataset to understand user rating behavior and derive actionable insights for recommendation systems. The dataset was cleaned, refined and enhanced with engineered features such as release year, top genre, movie age and average ratings. These findings provide a strong foundation for designing a personalized and accurate movie recommendation system.

Key insights include:

1. Most users rate movies positively, with ratings concentrated between 3.0 and 4.0.
2. Certain genres like Film-Noir and Mystery receive consistently higher ratings while Horror scores lower.
3. Older movies generally receive higher ratings than newer ones with a slight decline for movies older than 100 years.
4. User rating behavior is consistent regardless of the number of genres a movie has.

Feature Created

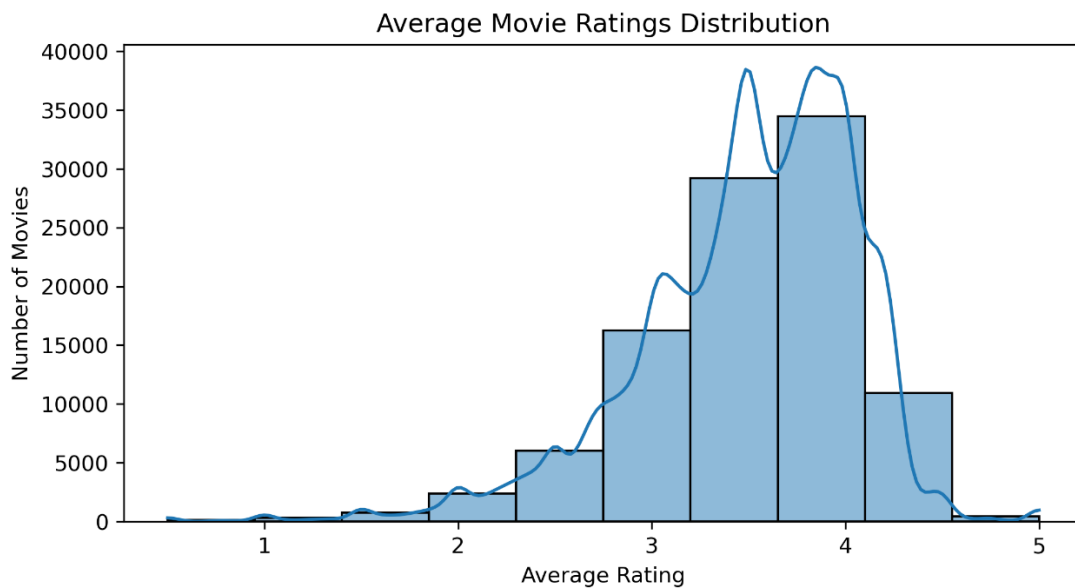
To improve insights and modeling, the following features were created:

1. Release Year: This was extracted from the title column to help analyze temporal bias and trends over time.
2. Number of genres: This is the count of genre in which a movie belongs to. It is used to analyze how number of genres a movie has affect its rating and user's preference
3. Top genre: This is the primary genre a movie belongs to. It was created so as to confirm if the main genre influences user preference or not.

4. Average user rating: This is the mean rating given by users across all their ratings. This is crucial for normalizing ratings.
5. Average movie rating: This is the mean rating of each movie across users. This helps in recommendation accuracy compared to rating, using average ratings helps reduce noise from individual rating variations and improves analysis accuracy.
6. Age of movie: This is the number of years since movie release. This makes comparative analysis and popularity trend over time possible.

Exploratory Data Analysis

1. Average Rating Distribution:

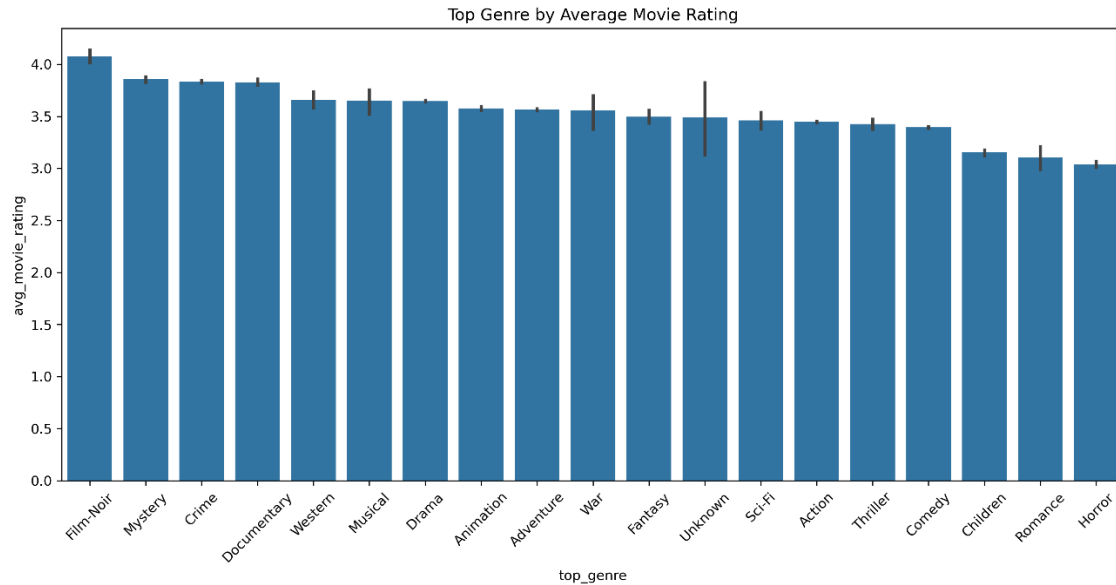


Most movies have a rating between 3.0 and 4.0 indicating that most users rate movies positively. Also, low number of movies receive low ratings or perfect ratings. This shows that users rating behavior is consistent and moderate.

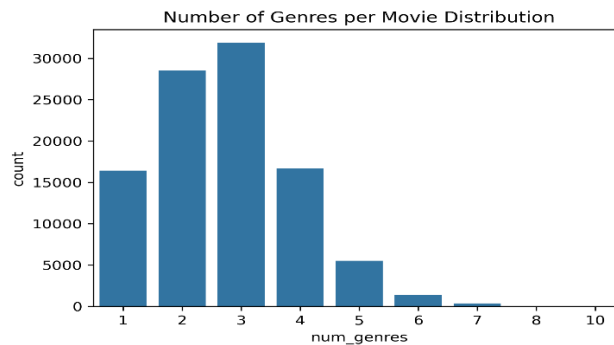
2. Top Genre Analysis:

Film-noir genre receive the highest average rating followed by mystery, crime and documentary while the genre horror received the least average rating.

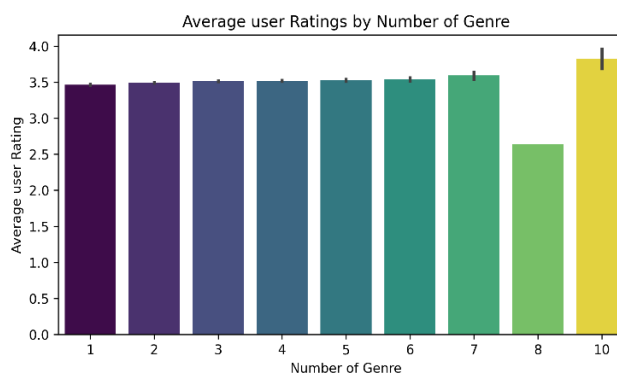
This indicates that most users tend to enjoy and rate movies with a clear, direct focus on good topics rather than frightening or scary movies. With this, when recommending movies for users, movies from highly rated genre should be recommended above horror.



3. Number of Genre:

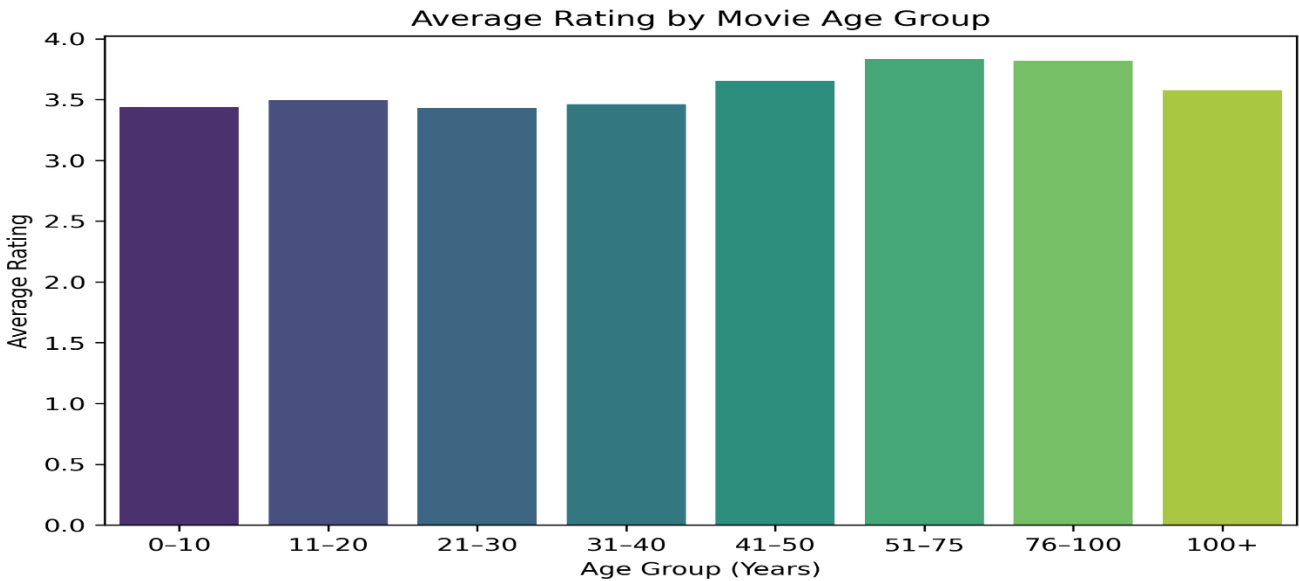


Most movies have ≤ 3 genres while very few have more than 6 genre which indicates most recommended movies have direct focus and not too broad. But does the number of genres affect users rating?



The chart here answered the previous question. The user rating is not affected by the number of genres as the user rating remains consistent and moderate regardless of the number of genres a movie has. This means the rating is based on movie quality and other factors.

4. Movie Age Analysis:



Movie rating gradually increase with age as seen in the chart. Older movies tend to receive slightly higher ratings than newer releases and a sharp decline in rating of overly old movies (>100 years). The sharp decline for movies older than 100 years may be attributed to limited viewership or accessibility. Overall, the chart suggests that audience prefer movies between 51 – 100 years old. So, movies within this category should be recommended the most for the audience.

5. Most Rated Movies

	title	avg_movie_rating	num_genres	age_of_movie
66992	Peaceful Warrior (2006)	5.0	1	19
16936	Loving Vincent (2017)	5.0	3	8
16885	The Girls (1961)	5.0	2	64
64142	Belle époque (1992)	5.0	2	33
17249	Rivers and Tides (2001)	5.0	1	24
28004	What Happened Was... (1994)	5.0	4	31
68366	Odd Life of Timothy Green, The (2012)	5.0	3	13
78207	Tyler Perry's I Can Do Bad All by Myself (2009)	5.0	2	16
16886	The Adventures of Sherlock Holmes and Doctor W...	5.0	1	2025
16917	Vacations in Prostokvashino (1980)	5.0	1	45

The table shows that most of the highly rated movies are classics and most have <4 number of genres. This suggests that number of genres also affects the ratings but the impact maybe insignificant as this is a very tiny piece of data given our large dataset. Also, higher rating may be as a result of other factor like producing company, quality amongst other reasons.

How the Insights could Support Building a recommendation System

The insights and features developed in this analysis can be directly applied to recommendation systems:

1. features such as average user rating and average movie rating can be used to measure similarity between users and movies enabling the system to suggest titles that users with similar preferences have enjoyed.
2. personalization can be enhanced by normalizing user ratings with the average user rating feature which helps reduce bias from users who consistently rate higher or lower than others.
3. features like top genre, number of genres and age of movie help build detailed movie profiles allowing recommendations based on a user's viewing history and preferred content characteristics.

Conclusion

The analysis revealed clear patterns in user and movie behavior. Feature engineering, particularly capturing movie age, genre count and user ratings can significantly enhance recommendation systems. Future models can also leverage these insights for both accuracy and relevance.